One-dimensional Kalman filters

Kalman and Bayesian Filters in Python, chapter 4

Lesson t2.2 · R. Labbe

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Why one number is worth a filter

- A drone's **barometer altitude** is one number with one peak of uncertainty
- A wheel's **encoder distance** drifts by a little every step
- A gyro **bias** is a slowly wandering scalar
- Each is a scalar Kalman filter in practice, and the matrix filters reuse every line

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Roadmap

1. Beliefs as Gaussians
2. Predict: sum of Gaussians
3. Update: product of Gaussians
4. The loop and the Kalman gain
5. Designing the filter: process variance
6. Steady state and fixed gain

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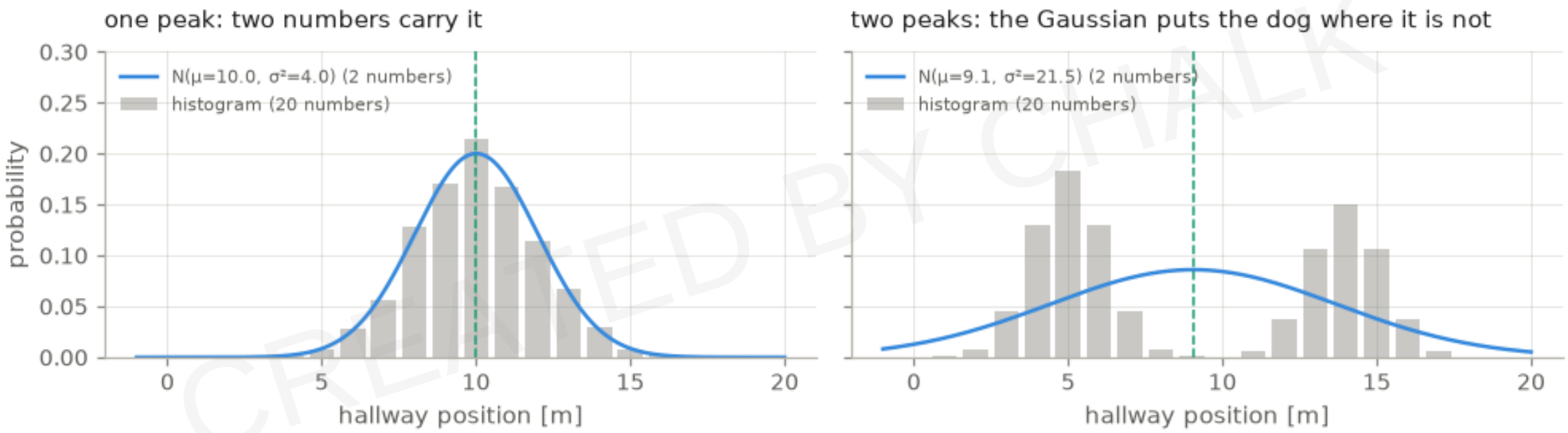
Part 1 of 4

Beliefs as Gaussians

Replace the histogram by a mean and a variance

A histogram belief and its Gaussian

Fig L1 · p. 134 A Gaussian replaces the histogram only when the belief has one peak



One peak: faithful. Two peaks: misleading

One peak needs two numbers; two peaks need a histogram or particles

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Predict: means add, variances add

$$\bar{x} = x + u, \qquad \bar{P} = P + R_{proc}$$

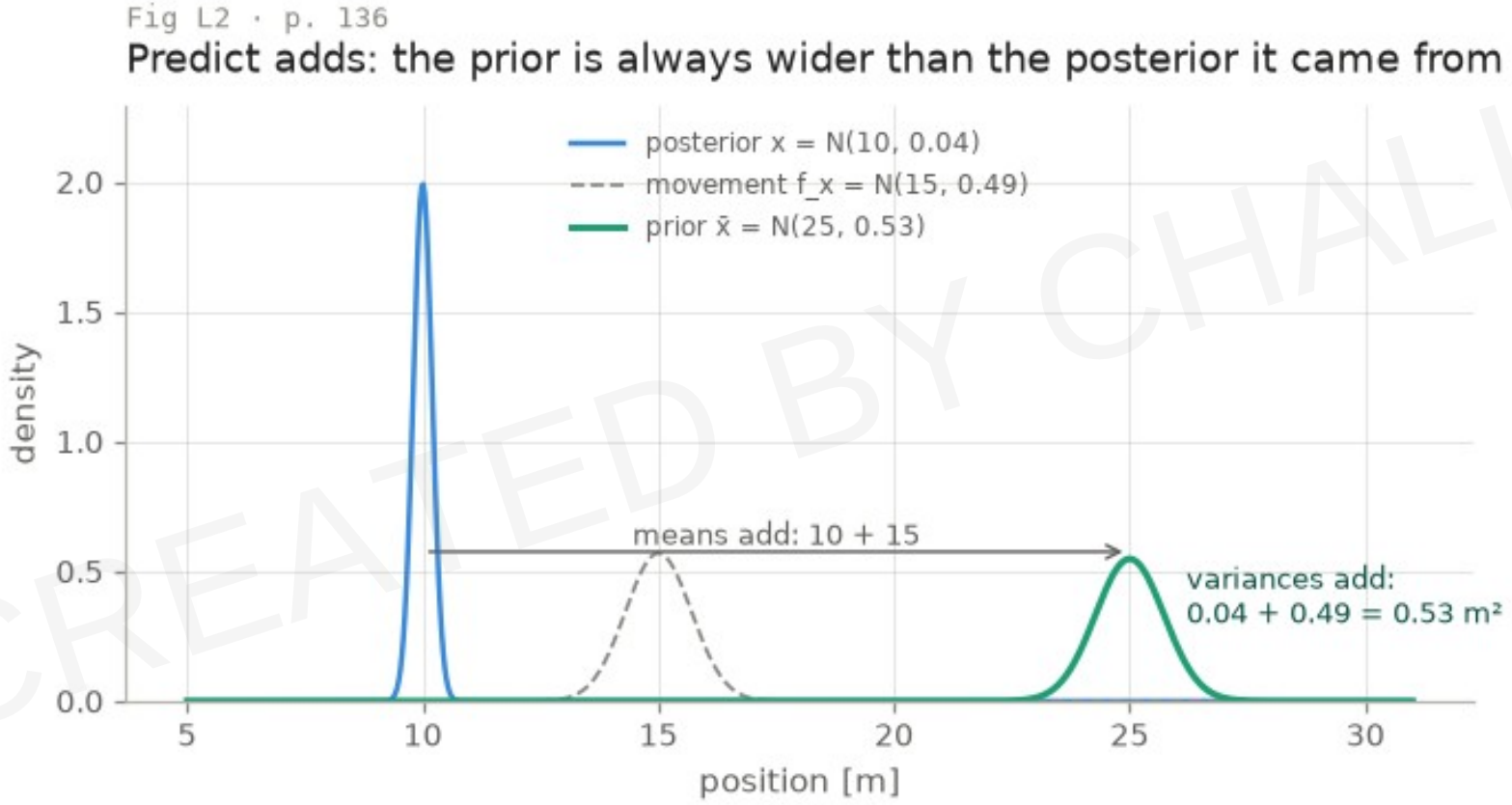
(p. 153)

- **$\bar{x}$, x** prior and posterior mean, m
- **$\bar{P}$, P** variance, m²
- **u** known displacement, m/step
- **R_proc** variance added per step, m²

Predict shifts the belief and widens it; it never narrows

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N(10, 0.04) + N(15, 0.49) = N(25, 0.53)



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Means add and variances add: the result is wider than either input

Predict: two predicts, no measurement

- Prior variance P
- Each predict adds R_proc

? **After two predicts and no update, what is the variance?**

A P + R_proc

B P + 2·R_proc

C P + R_proc^2

D P

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Update: the product of two Gaussians

$$\mu = \frac{\bar{\sigma}^2 \mu_z + \sigma_z^2 \bar{\mu}}{\bar{\sigma}^2 + \sigma_z^2}, \qquad \sigma^2 = \frac{\bar{\sigma}^2 \sigma_z^2}{\bar{\sigma}^2 + \sigma_z^2}$$

(p. 138)

$\bar{\mu}$, $\bar{\sigma}^2$ prior mean and variance

μ_z the measurement z

σ_z^2 Q_meas, sensor variance

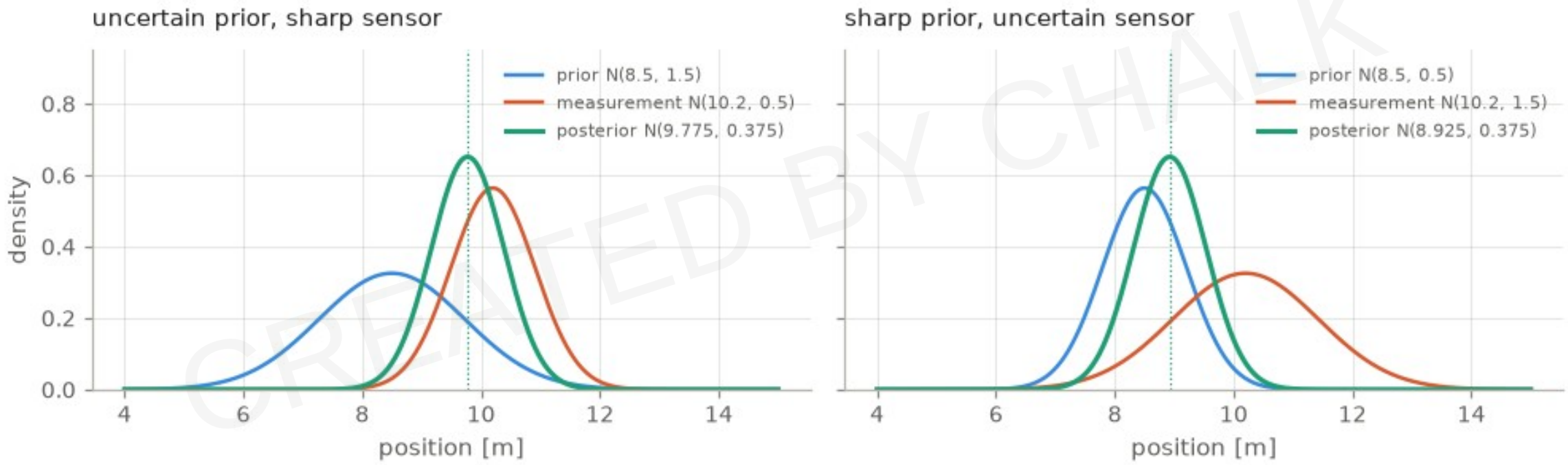
μ, σ^2 posterior mean and variance

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The variance ignores where the means are

The product is narrower than both inputs

Fig L3 · pp. 138–142 The product is narrower than both inputs and sits nearer the sharper one



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The posterior is 0.375 in both panels and sits near the sharper input

The one idea of the update

A measurement always reduces the variance, by an amount set by the two variances alone

Same code, a different outcome: decided by two variances, not by the data

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Part 2 of 4

The loop and the Kalman gain

Predict, update, repeat

The five-line filter of page 147

1 Predict mean

$\bar{x} = x + u$: move the belief by what we commanded

2 Predict variance

$\bar{P} = P + R_{proc}$: widen it by the model's doubt

3 Update mean

$x = (\bar{P}z + Q_{meas}\bar{x})/(\bar{P} + Q_{meas})$

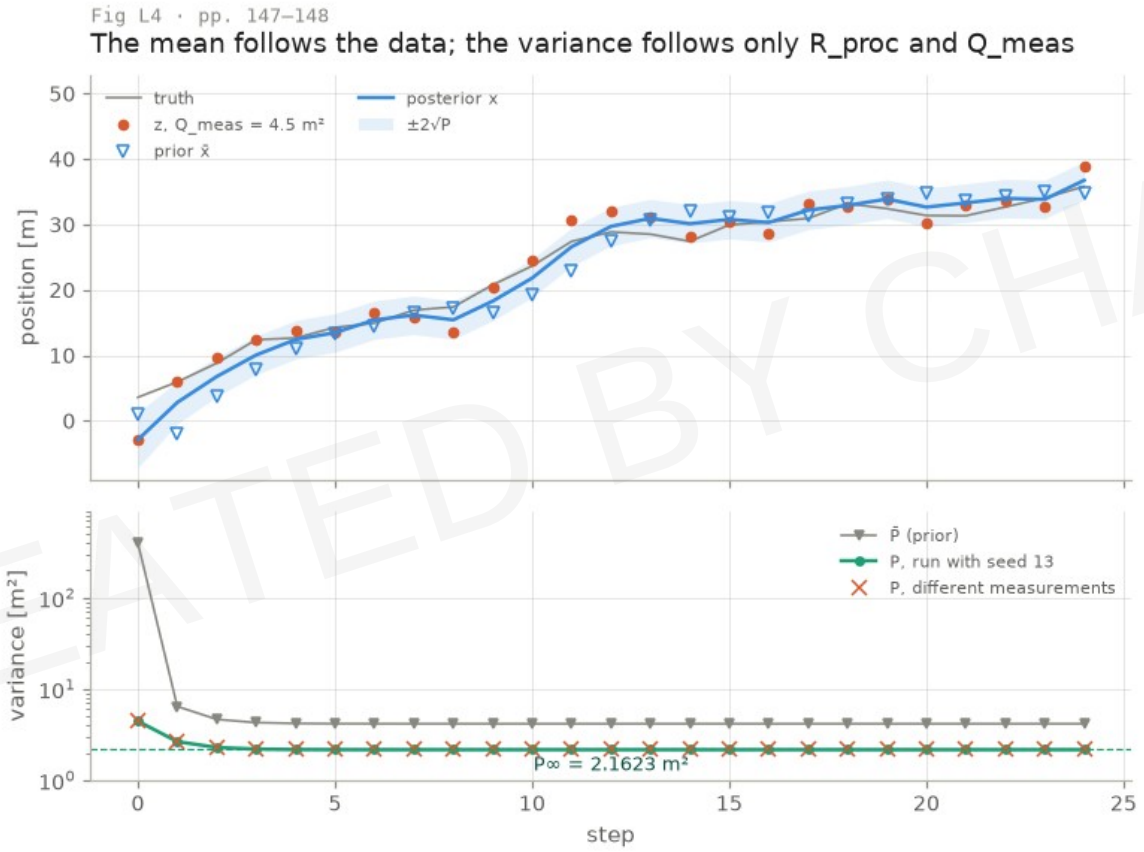
4 Update variance

$P = \bar{P}\,Q_{meas}/(\bar{P} + Q_{meas})$

Two lines predict, two lines update; the loop repeats per measurement

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The 25-step dog and its variance



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P is identical for two different measurement sets and settles at 2.1623 m²

The first step, measured

401

first prior variance, m²

1.990

1

first posterior variance, m²

1.000

0

P at step 10, R_proc = 1, Q_meas = 2

2.162

3

P at step 9 for (2, 4.5), m²

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Kalman gain form of the update

$$y = z - \bar{x}, \quad K = \frac{\bar{P}}{\bar{P} + Q_{meas}}, \quad x = \bar{x} + Ky, \quad P = (1 - K)\bar{P}$$ (p. 153)

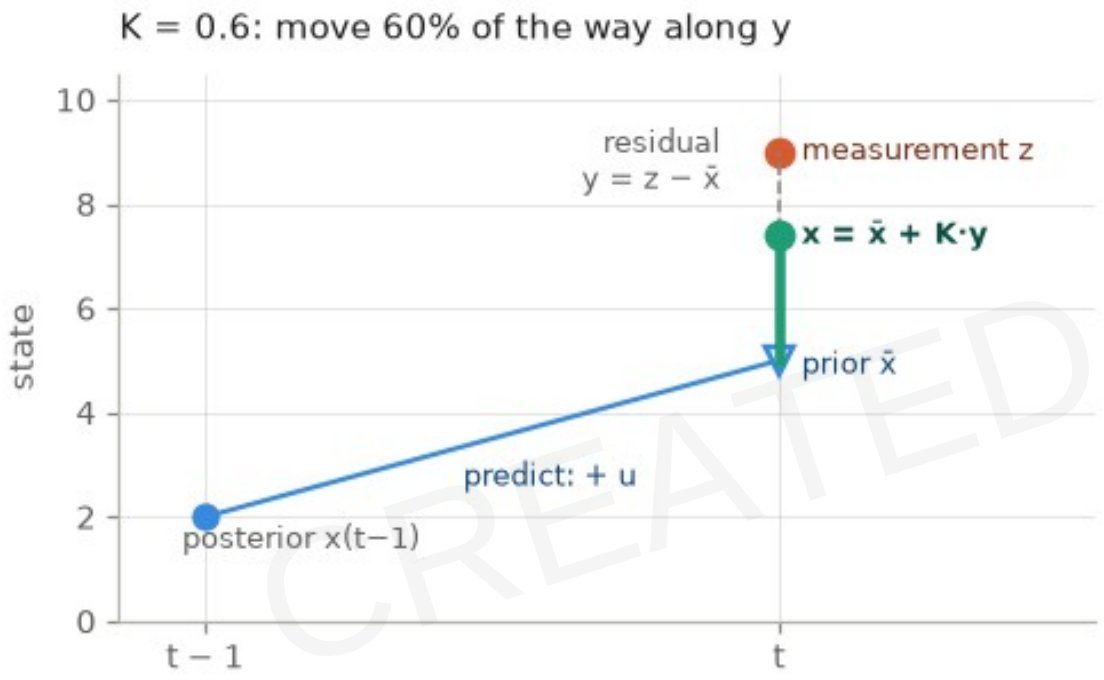
- **y** residual, m
- **K** gain, between 0 and 1
- **P̄** prior variance, m²
- **Q_meas** sensor variance, m²

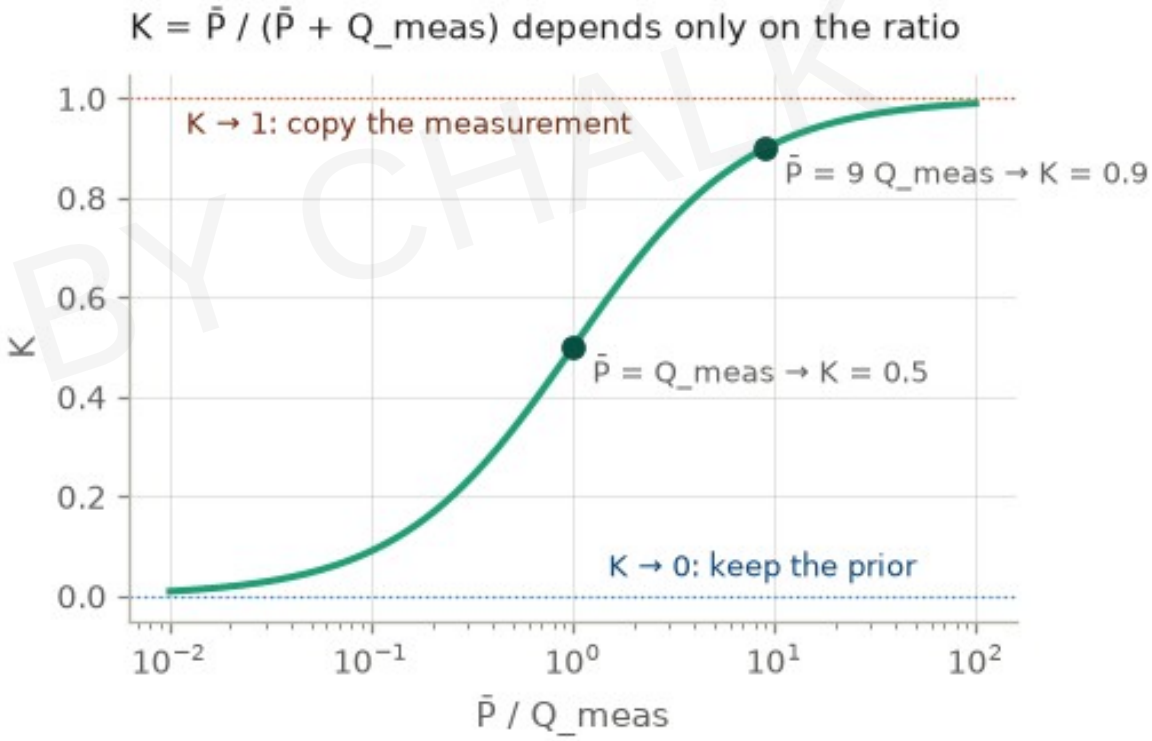
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Prior plus a fraction of the surprise: the g-h filter with a computed gain

Residual and gain

Fig L5 · pp. 149–151 The gain picks a point on the residual; the variance ratio picks the gain





K depends only on the ratio of the prior variance to the sensor variance

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Predict: the gain for a wide prior

- Prior variance is 9 times the sensor variance

? **What is the Kalman gain K?**

A 1/10

B 1/2

C 9/10

D 9

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Three ways to model a sensor's error

g-h filter

- Gain picked by hand
- Uniform error bars

Discrete Bayes

- Any histogram
- Models anything, resists analysis

Kalman filter

- Gaussian, closed under sum and product
- The variances compute the gain

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Every filter in the book is this recipe with different math in the boxes

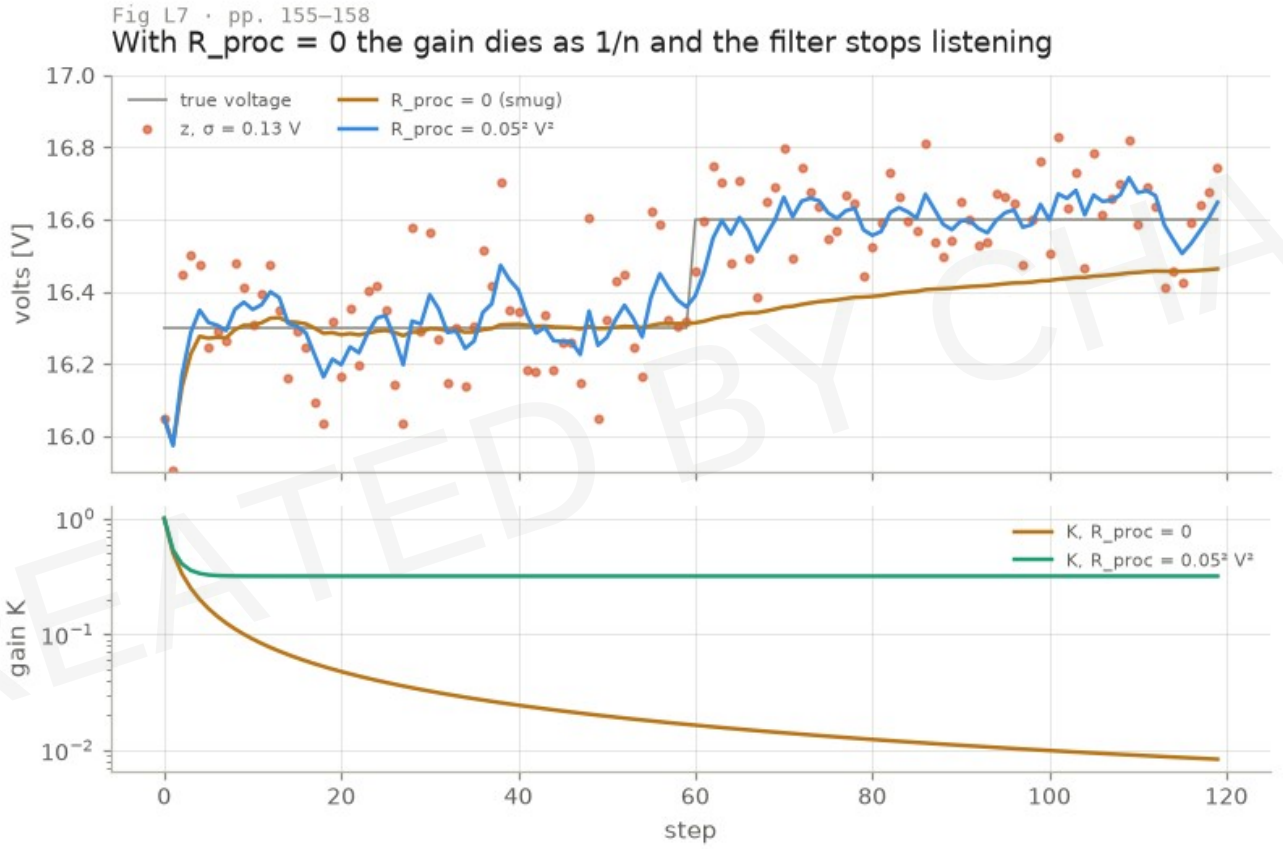
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Part 3 of 4

Designing the filter

What R_proc and Q_meas do

The smug filter: R_proc = 0



No movement means u = 0, not R_proc = 0

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Right model, wrong model

15.06

RMS error in m, σ_z = 300 m, right model

292.8

RMS error in m of the raw measurement

342.1

mean NIS, R_proc = 0.001, wrong model

1.82

mean NIS, R_proc = 10

- NIS = y^2/S, $S = \bar{P} + Q_meas$, mean 1 for a consistent scalar filter

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Predict: a bad starting estimate

- x_0 is 1000 m off, $P_0 = 500$
- Q_meas = 25 m², R_proc = 2 m²

? **After the first update, roughly how large is the error?**

A About 0 m

B About 47 m

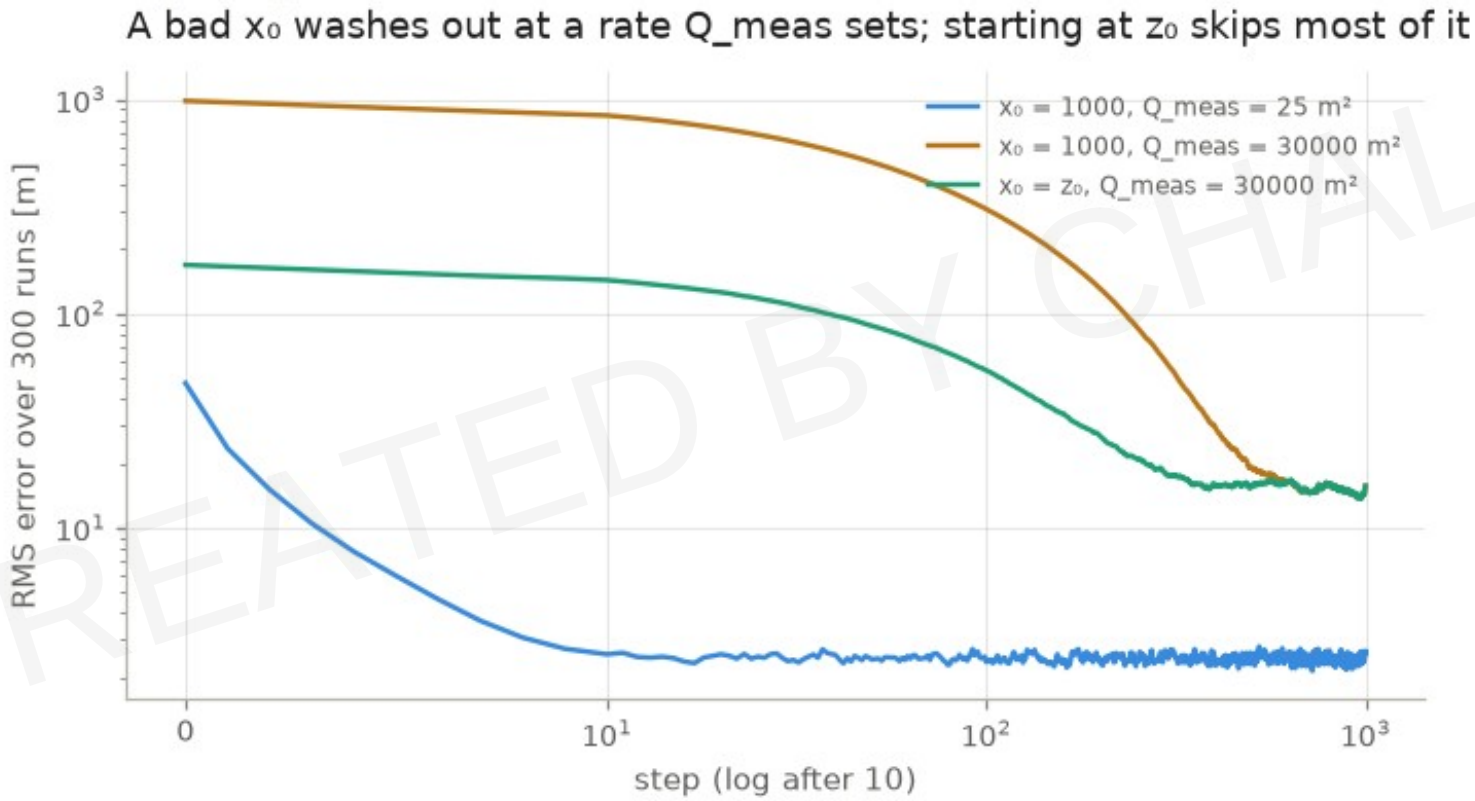
C About 500 m

D About 1000 m

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Initial error washes out at a rate set by Q_meas

Fig L9 · pp. 161–164



Initialize from the first measurement, with P_0 equal to that sensor's variance

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Part 4 of 4

Steady state and fixed gain

When the filter stops changing

The fixed point of the variance

$$P_\infty^2 + R_{proc}P_\infty - R_{proc}Q_{meas} = 0, \qquad K_\infty = \frac{P_\infty}{Q_{meas}}$$

(not in the chapter)

P∞ steady-state posterior variance, m²

K∞ steady-state gain

R_proc process variance per step

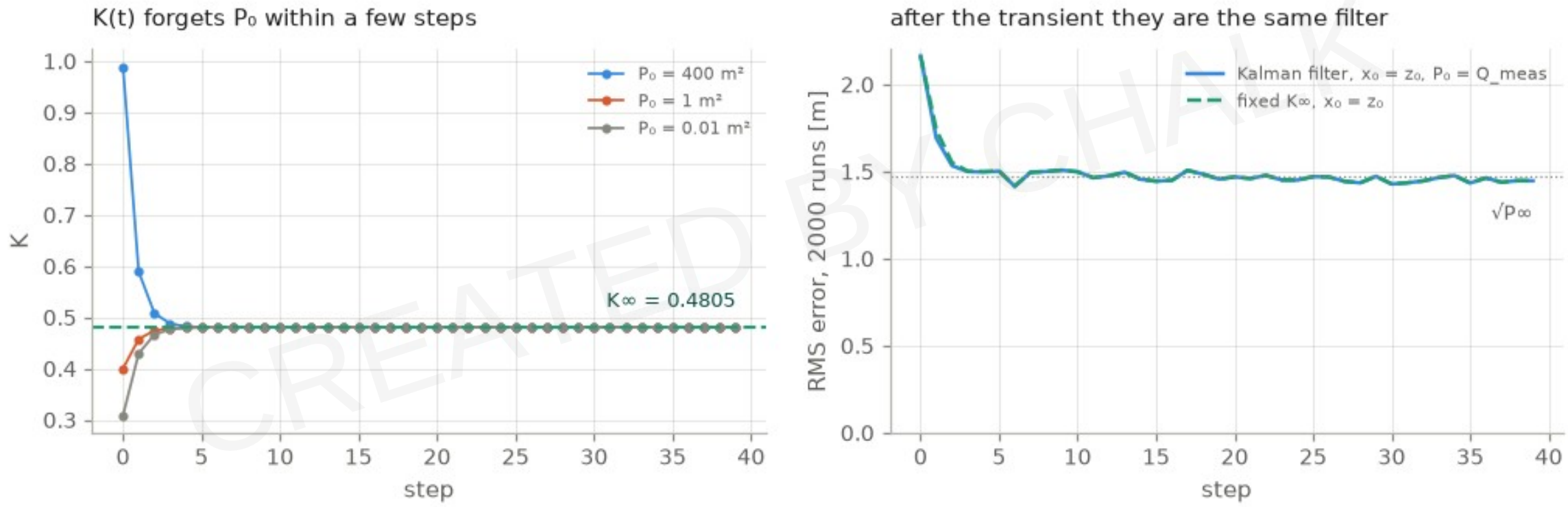
Q_meas sensor variance

Constant R_proc and Q_meas: P and K converge, from any P_0 above 0

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A hard-coded gain loses only the transient

Fig L10 · pp. 167–168 Constant R_proc and Q_meas make P and K constant; hard-code K∞



Starting at z_0, the Kalman and the fixed-gain filter are at 1.455 m RMS

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Symptoms and fixes

Symptom	Likely cause	Try
Estimate drifts off the data; NIS mean $\gg 1$	R_proc too small	Raise R_proc until NIS ≈ 1
K decays toward 0	R_proc = 0 (smug)	Small nonzero R_proc
As noisy as z; NIS mean $\ll 1$	R_proc large or Q_meas small	Measure Q_meas at rest
Constant offset that never decays	Biased u	Fix u or add a bias state
Lags a moving signal	No dynamics for the motion	Larger R_proc or velocity state
Long transient after start	Bad x_0, large Q_meas	$x_0 = z_0$, P_0 = Q_meas
Off by about a square	σ used as a variance	Square σ; name vars *_var

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Claims you should be able to defend

- Means add and variances add in predict, so predict never narrows
- The posterior variance is smaller than both inputs and ignores the means
- $K = \bar{P}/(\bar{P} + Q_meas)$ turns the update into prior plus a fraction of the residual
- R_proc = 0 makes the filter smug; the right R_proc is the model's honest doubt
- P converges, so a fixed gain K_∞ is a valid hard-coded filter

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What comes next

- Labbe ch. 6: x and P become a vector and a matrix
- Thrun ch. 3, Table 3.1: this chapter with scalars replaced by matrices
- Labbe ch. 8: choosing the process noise, which today only showed going wrong

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